HEM PATEL

Aspiring Machine Learning / AI / Data Science Engineer
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EDUCATION

New L. J. Institute of Technology — B.E., Computer Science & Engineering — Expected 2028

Indian Institute of Technology (IIT) Mandi — Minor Degree, Artificial Intelligence & Data Science — Expected 2026

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Solidity
AI / Machine Learning: TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, OpenAI Whisper, Computer Vision, NLP, LLM Evaluation, Prompt Engineering
Generative AI Tools: Claude Code, ChatGPT / GPT-4, Google Gemini
Frontend: React, React Native, SvelteKit, Tailwind CSS, HTML/CSS
Backend: FastAPI, Node.js, Express.js, REST APIs, Socket.IO, JWT Authentication
Tools & Platforms: Git/GitHub, GitHub Actions, CI/CD, pytest, mypy, Docker, SQLite, MongoDB, Vercel, ethers.js, VS Code

AI EVALUATION EXPERIENCE

Benchmark Task Author, AI Agent Evaluation (Handshake AI, Harbor framework) — 2026

- Authored Docker-packaged Harbor tasks in scientific computing (Hodgkin-Huxley, Wilson-Cowan, thermoacoustic Hopf oscillator, fault-cascade model), each with hidden-ground-truth data and a pytest verifier.
- Designed trap mechanisms and measured them: on the Hodgkin-Huxley task a naive Euler fit misses the forecast tolerance (0.19-2.26 mV RMSE vs 0.12 mV) while an adaptive solver passes (0.0016-0.060 mV).
- Validated verifiers locally with Harbor (reference solution scores 1.0, do-nothing agent 0.0) and added re-simulation checks to block fabricated answers.

PROJECTS

Vakil — Autonomous Chargeback Defence Agent
Python, scikit-learn, Docker, GitHub Actions, Razorpay API

- Built an eight-stage agentic pipeline (ingest, triage, evidence harvest, compelling-evidence check, decision, draft, file, ledger) that decides whether a merchant should contest a payment dispute and assembles the evidence pack. Built for the Razorpay AI Buildathon 2026, Track 2.
- Designed a Fight-or-Fold expected-value engine that prices each dispute against a break-even win probability, backed by a Platt-calibrated model selected via 5-fold cross-validation, reaching 0.789 F1 and 0.175 Brier score on a frozen held-out split.
- Implemented a provenance gate that resolves every factual sentence in the generated rebuttal against a closed fact index and removes anything unverifiable, so the filed letter cannot contain a fabricated claim.
- Engineered a hash-chained audit ledger with replay and tamper detection, backed by 196 tests, mypy strict typing, and CI that regenerates all evaluation metrics on every push.

RAG QA Chatbot — Data Structures Study Assistant
Python, FAISS, Hugging Face

- Built a fully local Retrieval-Augmented Generation pipeline that parses, indexes, and answers questions from technical PDF documents using a 100% free-tier stack.
- Implemented local embeddings (all-MiniLM-L6-v2) with a disk-persisted FAISS index, and used the Hugging Face Inference API (Gemma-2-9B-it) with grounding constraints so the model answers only from retrieved context or refuses.
- Built an LLM-as-judge harness that scores answer faithfulness on a 1–5 scale.

Iris Species Predictor & MLOps Pipeline
Python, scikit-learn, Flask, Docker, GitHub Actions

- Built a production-ready MLOps pipeline with a RandomForest classifier (~96%+ test accuracy) served via a Flask REST API with /health and /predict endpoints on Gunicorn.
- Automated CI/CD with GitHub Actions (tests, Docker build, in-container retraining, push to GitHub Container Registry) and set up one-click Render deployment with an interactive slider-based frontend.

Movie Recommendation System
Python, FastAPI, Docker, Pandas

- Built an item-based collaborative filtering recommender on MovieLens 100k using vectorized cosine similarity.
- Evaluated with Precision@10, served recommendations through a Dockerized FastAPI service with genre filtering and a cached model, and built a lightweight web UI.

PostPilot — AI LinkedIn Post Agent
Python, FastAPI, PostgreSQL, SQLAlchemy, Gemini API, Groq API, Render

- Built a multi-user web app that turns a topic into a LinkedIn post with hashtags and an image, then publishes it immediately or on a schedule, using Gemini with a Groq fallback and per-user daily limits.
- Built a scheduling worker on Postgres using SELECT FOR UPDATE SKIP LOCKED so concurrent workers never publish the same post twice, deployed on Render with Alembic migrations on every deploy.
- Secured the app with CSRF protection, Fernet-encrypted LinkedIn tokens, per-route ownership checks, and full account export and deletion, tested with pytest in an isolated database schema.

CERTIFICATIONS & ACHIEVEMENTS

Certificate of Excellence in Evaluation — Minor in AI & Data Science, CCE, IIT Mandi (in collaboration with Masai) — August 2026

Finalist, The MaverickEffect AI Challenge — Dewang Mehta Foundation — August 2026

Participant, Ifest'25 — Dhirubhai Ambani University — October 2025

1st Runner-Up, RotaHack 2025 — Rotaract Club of New L.J.I.E.T — August 2025

Campus Partner, Perplexity AI — Aug 2025 – Nov 2025